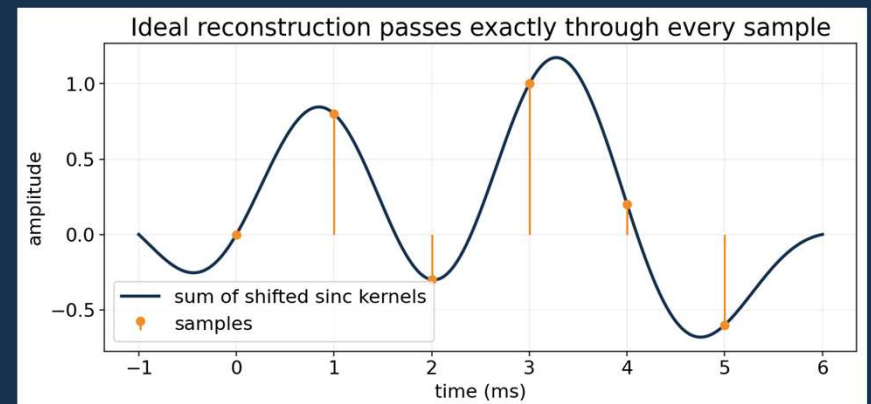


Reconstruction, D/A, and Quantization Noise

Ideal interpolation gives the theory; ZOH and finite bit depth explain what hardware actually does.



75 minutes • ideal D/C • sinc interpolation • zero-order hold • quantization • SQNR • PyTorch dtype bridge

Recap: after Lecture 3, we trust the sample times

Now we ask how to reconstruct between samples and represent amplitudes.

Sampling error

Where Week 2 lives in the audio → ML pipeline

Quantization error



Aliasing moves energy to the wrong frequency.
It is fundamentally non-unique.

Quantization maps a continuous amplitude to
one of finitely many levels.

Learning outcomes

By the end of Lecture 4, students should be able to move between ideal theory and real hardware.

- Derive the ideal bandlimited interpolation formula from the reconstruction low-pass filter.
- Explain zero-order hold behavior in time and its sinc-shaped magnitude droop.
- Model a uniform B-bit quantizer and compute Δ and quantization-error variance.
- Derive and use $\text{SQNR} \approx 6.02B + 1.76 \text{ dB}$ for a full-scale sine.
- Explain the difference between 16-bit PCM storage and float32 computation in PyTorch.

Ideal reconstruction in the frequency domain

If spectral replicas do not overlap, simply isolate the baseband copy.

$$X_s(j\Omega) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s))$$

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| < \Omega_c \\ 0, & \text{otherwise} \end{cases}$$

The factor T cancels the 1/T introduced by ideal sampling.

Choose $\Omega_M < \Omega_c < \Omega_s - \Omega_M$ so only one replica passes.

From ideal low-pass filtering to sinc interpolation

The inverse transform of a brick-wall low-pass filter is a sinc.

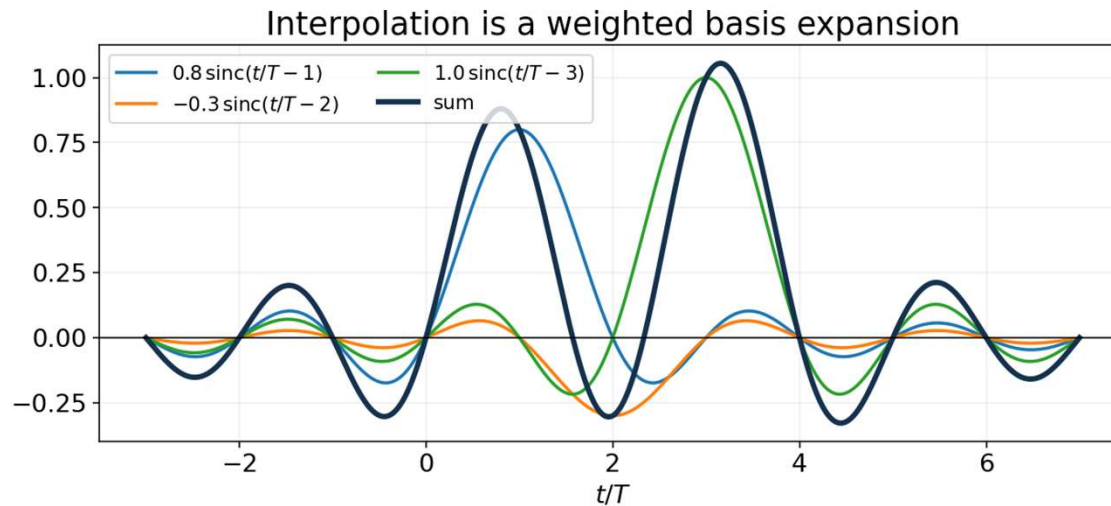
$$h_r(t) = \text{sinc}\left(\frac{t}{T}\right) \quad \text{for the ideal Nyquist-band reconstruction case}$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t - nT}{T}\right)$$

Here $\text{sinc}(u) = \sin(\pi u) / (\pi u)$. Every sample contributes one shifted, scaled interpolation kernel.

Why the sinc basis passes through every sample

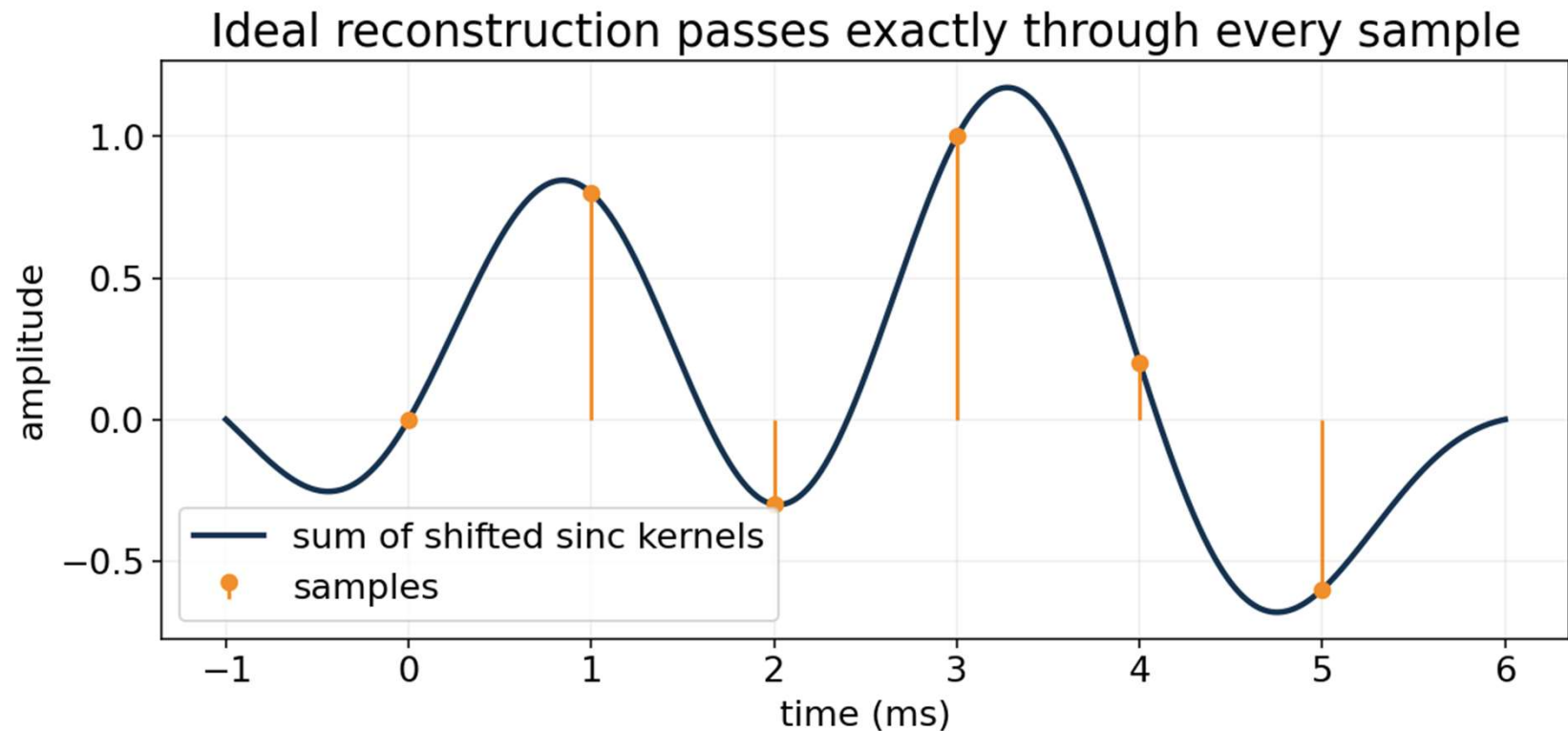
At integer offsets, normalized sinc is zero—except at the center.



At $t=mT$, only the $n=m$ kernel survives $\rightarrow x_r(mT)=x[m]$.

A finite illustration of ideal interpolation

The true ideal sum is infinite; a finite example still shows the geometry.



Why a real DAC does not implement ideal sinc directly

The ideal impulse response is infinite in duration and noncausal when centered at $t=0$.

Ideal mathematical interpolator

- Infinite support in time.
- Symmetric around the interpolation instant.
- Brick-wall frequency response.
- Perfect only under ideal bandlimiting assumptions.

Practical hardware strategy

- Generate a simple analog waveform from each code.
- Zero-order hold is extremely natural in a DAC.
- Follow with an analog reconstruction / anti-imaging filter.
- Oversampling reduces the burden on analog filtering.

Ideal theory tells us the target; implementation approximates it.

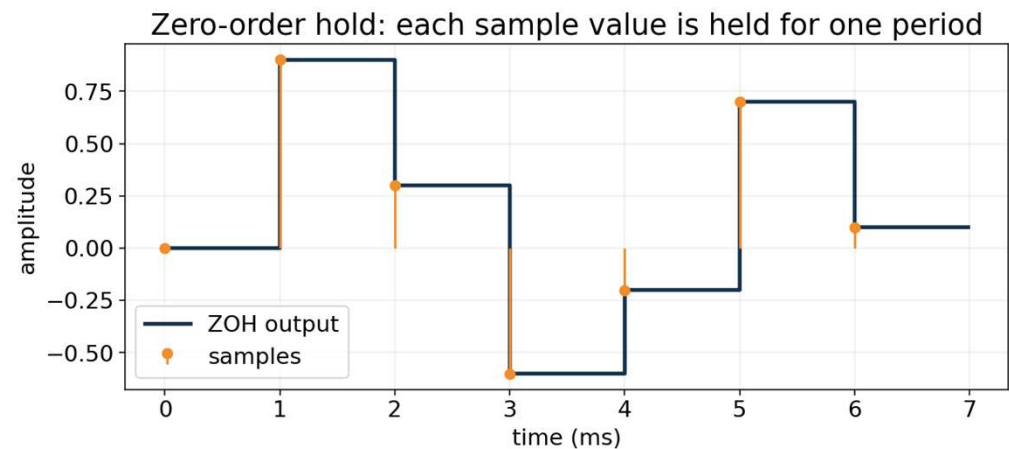
Zero-order hold (ZOH): hold each sample for one period

A rectangular pulse is convolved with the weighted impulse train.

$$h_{ZOH}(t) = u(t) - u(t - T)$$

$$x_{ZOH}(t) = x_s(t) * h_{ZOH}(t)$$

This staircase is already a continuous-time waveform—but it contains spectral images and amplitude droop.



Deriving the ZOH frequency response

Transform the rectangular hold pulse.

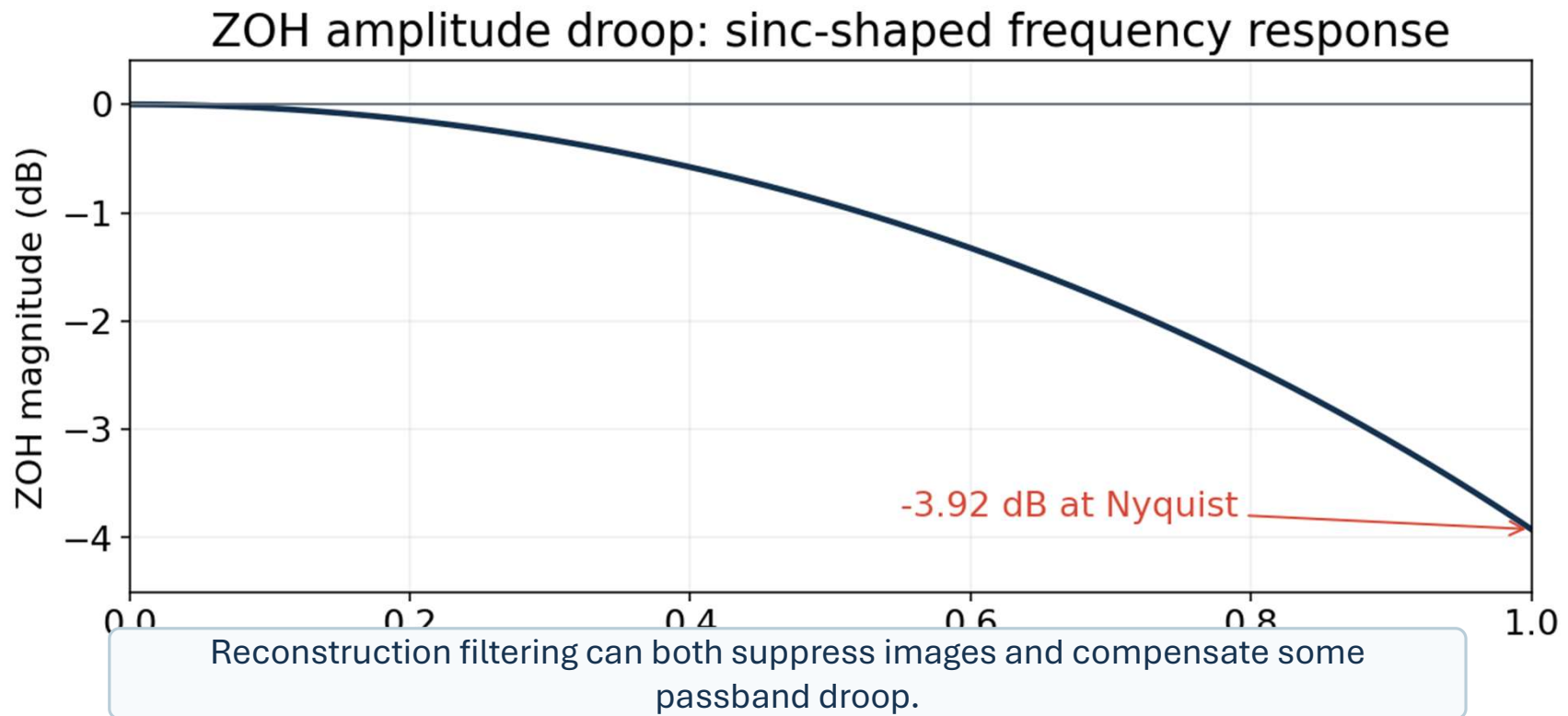
$$\begin{aligned} H_{ZOH}(j\Omega) &= \int_0^T e^{-j\Omega t} dt \\ &= T e^{-j\Omega T/2} \frac{\sin(\Omega T/2)}{\Omega T/2} \end{aligned}$$

- Linear phase term: $e^{-j\Omega T/2}$ corresponds to a $T/2$ delay.
- Magnitude term: sinc-shaped envelope.
- High frequencies are attenuated more than low frequencies.

At $\Omega T = \pi$: $|H|/T = 2/\pi \approx 0.637 \rightarrow -3.92$ dB.

The ZOH creates sinc-shaped amplitude droop

This is deterministic frequency response—not random noise.



A practical D/A chain

ZOH is only one stage; analog filtering completes the reconstruction.



The analog LPF removes spectral images around multiples of f_s ; oversampling gives the filter more transition width.

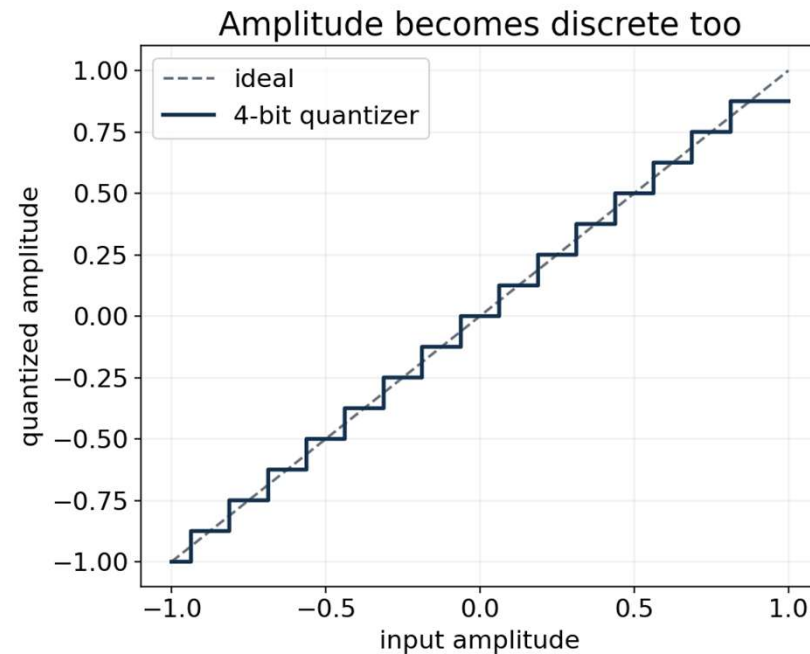
Quantization: the second discretization

Sampling discretizes time. Quantization discretizes amplitude.

$$x_q[n] = Q\{x[n]\} = x[n] + e[n]$$

$$\Delta = \frac{2A}{2^B}$$

- B bits $\rightarrow 2^B$ nominal levels.
- Δ is the quantizer step for full-scale range roughly $[-A, A]$.
- Overload/clipping is a separate failure mode when $|x|$ exceeds the designed range.



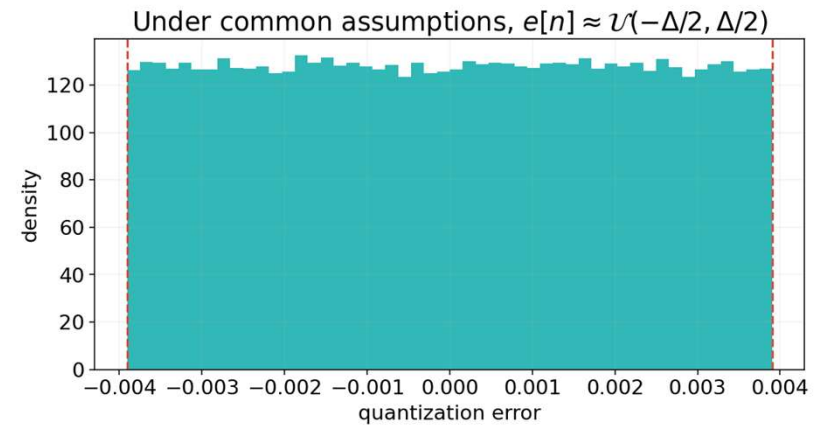
The additive quantization-noise model

Useful—but only under assumptions.

$$e[n] = x_q[n] - x[n]$$

$$e[n] \approx \mathcal{U}(-\Delta/2, \Delta/2)$$

$$\mathbb{E}\{e[n]\} \approx 0, \quad \sigma_e^2 \approx \frac{\Delta^2}{12}$$



Good approximation when the signal spans many levels and the error is sufficiently decorrelated from the signal.

Derive the quantization-noise variance

Uniform error over one quantization interval.

$$p_e(e) = \frac{1}{\Delta}, \quad -\Delta/2 \leq e < \Delta/2$$

$$\sigma_e^2 = \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta} de = \frac{\Delta^2}{12}$$

Every additional bit halves $\Delta \rightarrow$ noise power falls by a factor of 4 $\rightarrow$ about 6 dB improvement.

Full-scale sine SQNR: where $6.02B + 1.76$ dB comes from

This is not an arbitrary rule of thumb; it follows from signal and error power.

$$P_x = \frac{A^2}{2}, \quad P_e = \frac{\Delta^2}{12}, \quad \Delta = \frac{2A}{2^B}$$

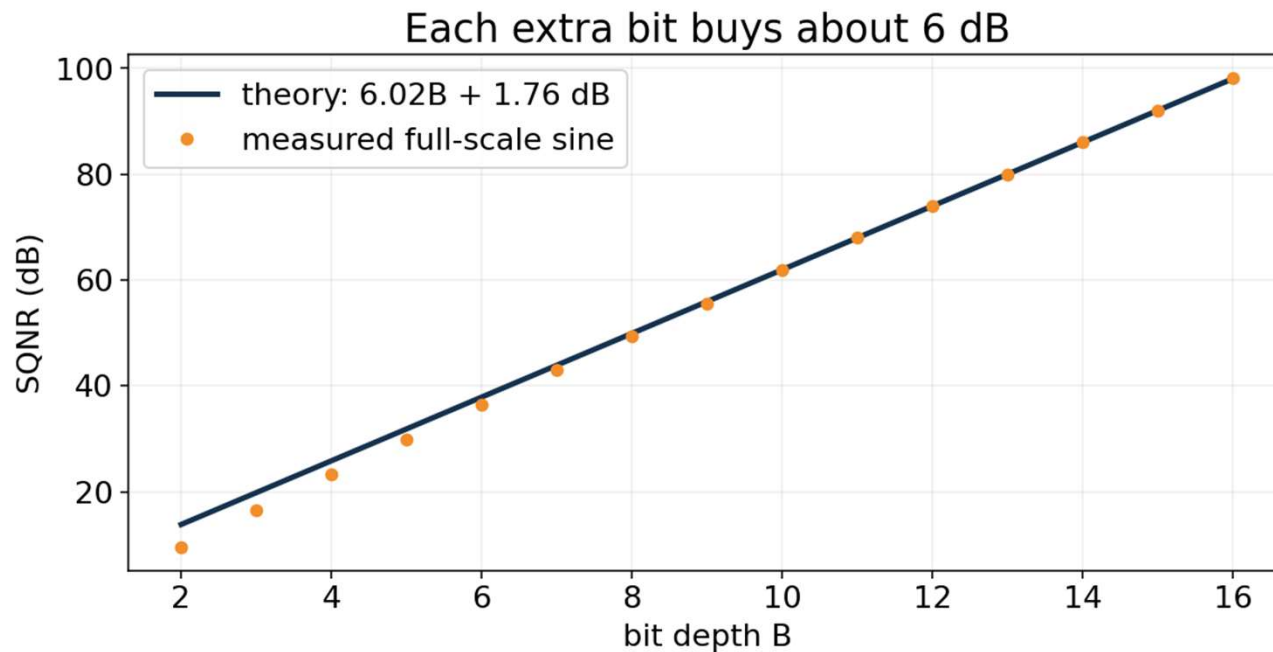
$$\text{SQNR} = 10 \log_{10} \frac{P_x}{P_e} = 10 \log_{10} \left(\frac{3}{2} 2^{2B} \right)$$

$$\text{SQNR} \approx 6.02B + 1.76 \text{ dB}$$

The +1.76 dB constant is specific to a full-scale sine and the ideal uniform-quantizer model.

About 6 dB per bit—verified numerically

The simple model is remarkably accurate at moderate/high resolution.



4-bit results deviate more because the high-resolution assumptions are weak.

16-bit full-scale sine theory: 98.08 dB SQNR.

16-bit audio: “96 dB” and “98 dB” can both appear

They refer to slightly different reference calculations.

6 dB/bit rule of thumb

$$20 \log_{10}(2^{16}) \approx 96.33 \text{ dB}$$

96.3 dB is the amplitude ratio across 16 binary bits.

Full-scale sine SQNR

$$6.02(16) + 1.76 \approx 98.08 \text{ dB}$$

98.1 dB includes the RMS power of a full-scale sine and uniform error model.

Be precise about what quantity is being compared.

PCM storage is not the same thing as ML compute dtype

Casting to float changes representation—not the information already captured.

```
pcm = np.array(..., dtype=np.int16)
x = pcm.astype(np.float32) / 32768.0
xt = torch.from_numpy(x)
# xt.dtype == torch.float32
```

- 16-bit PCM: discrete integer codes from the recording/ADC pipeline.
- float32 tensor: convenient numerical format for DSP and neural-network arithmetic.
- Converting int16 $\rightarrow$ float32 cannot recreate amplitude values between the original quantization levels.
- A float32 tensor can still contain badly aliased or coarsely quantized data.

More bits in the variable $\neq$ more bits of information in the measurement.

Live audio demo: listen in this order

The package includes pre-generated WAV files and notebook code.

- 1 **Clean 48 kHz** reference
- 2 **Naive 48→12 kHz** hear aliasing
- 3 **Filtered 48→12 kHz** compare controlled bandwidth loss
- 4 **8-bit quantized** notice noise floor
- 5 **4-bit quantized** strong granular/fuzzy error

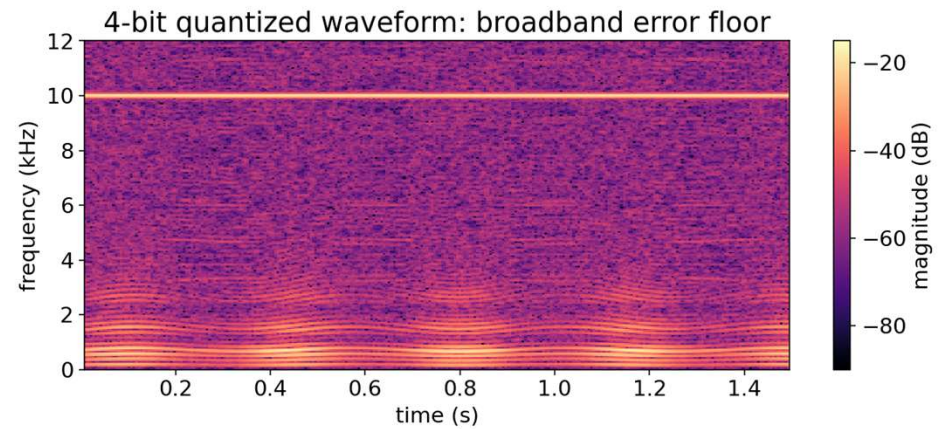
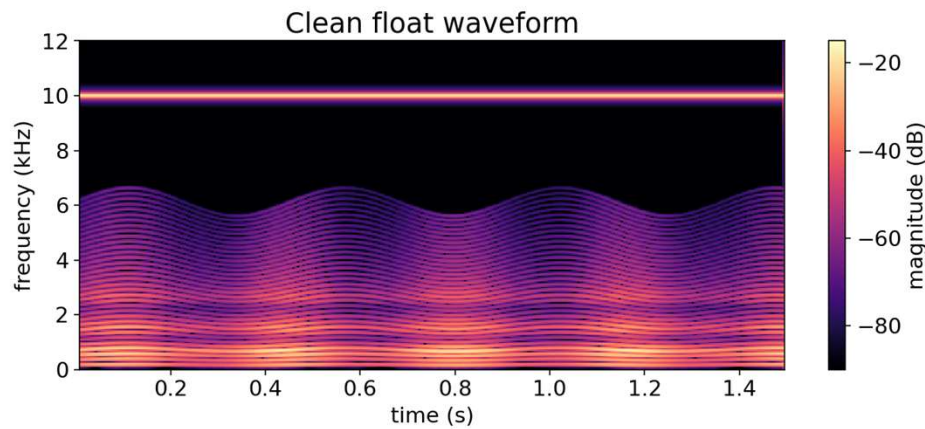
Ask students to predict what changes before playing each file.

What 4-bit quantization does to a spectrogram

The error is often broad-band, but it need not be perfectly white or independent.

Clean

4-bit



Graduate nuance: when the “white quantization noise” story breaks

Dither is one tool for trading correlated distortion for more noise-like error.

- Low-level sinusoids can produce periodic, signal-correlated error rather than white noise.
- Overload clipping is nonlinear distortion and is not captured by $\Delta^2/12$.
- Dither intentionally adds noise before quantization to decorrelate the quantization error.
- Noise shaping can push quantization noise toward frequency regions that matter less—but requires feedback/oversampling ideas beyond today.

**Modeling lesson for ML:
know which “noise
assumptions” are physics
and which are convenient
approximations.**

Exit check

Students answer individually; collect or discuss in the last two minutes.

1 Why does ideal reconstruction use sinc functions?

2 What deterministic frequency-domain effect does a ZOH introduce?

3 A full-scale 12-bit sine has approximately what ideal SQNR?

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Students answer individually; collect or discuss in the last two minutes.

1 Why does ideal reconstruction use sinc functions?

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Answers: 1) sinc is the impulse response of the ideal bandlimiting reconstruction filter. 2) sinc-shaped magnitude droop (plus linear phase/delay). 3) ≈ 74 dB.

Lecture 4 summary: faithful digital audio needs both time and amplitude integrity

Sampling theory, reconstruction, and quantization are one end-to-end system.

- Ideal D/C isolates one spectrum replica and yields sinc interpolation.
- A zero-order hold is practical but imposes sinc-shaped passband droop and spectral images.
- A B-bit uniform quantizer has step $\Delta \approx 2A/2^B$; under standard assumptions $\sigma_e^2 \approx \Delta^2/12$.
- A full-scale sine gives $\text{SQNR} \approx 6.02B + 1.76$ dB.
- `int16` $\rightarrow$ `float32` is a computational conversion; it does not undo aliasing or quantization.

Week 2 principle:
Bad acquisition creates
“features” that no
downstream ML method
can reliably repair.